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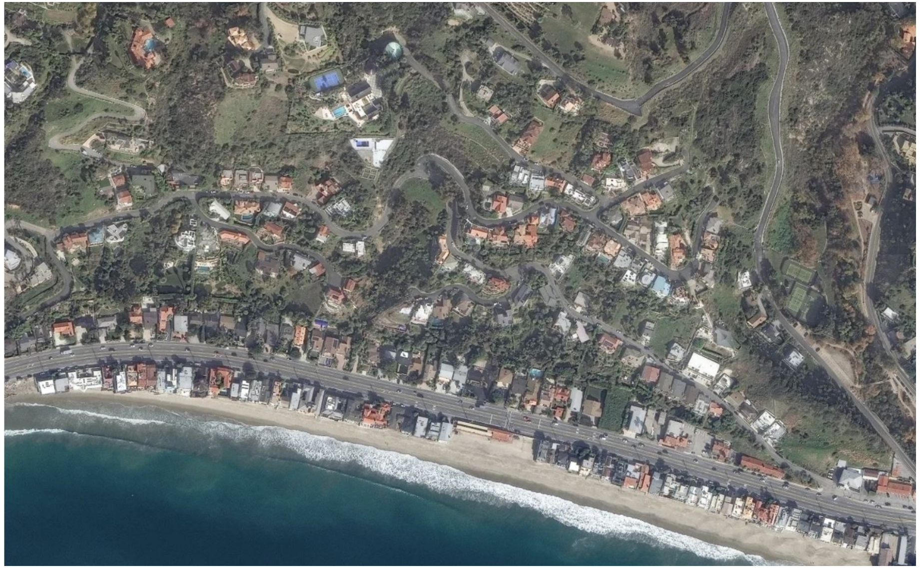
Fire-Flight
ML-Driven Aerial Sensing for Wildfire Prediction and Smart Evacuation

Milana Rubin, Rys Corpuz, Luise Haller
On behalf of Foothill Engineering Club's RC Plane Team

1. Wildfire Prediction: Strengths & Gaps

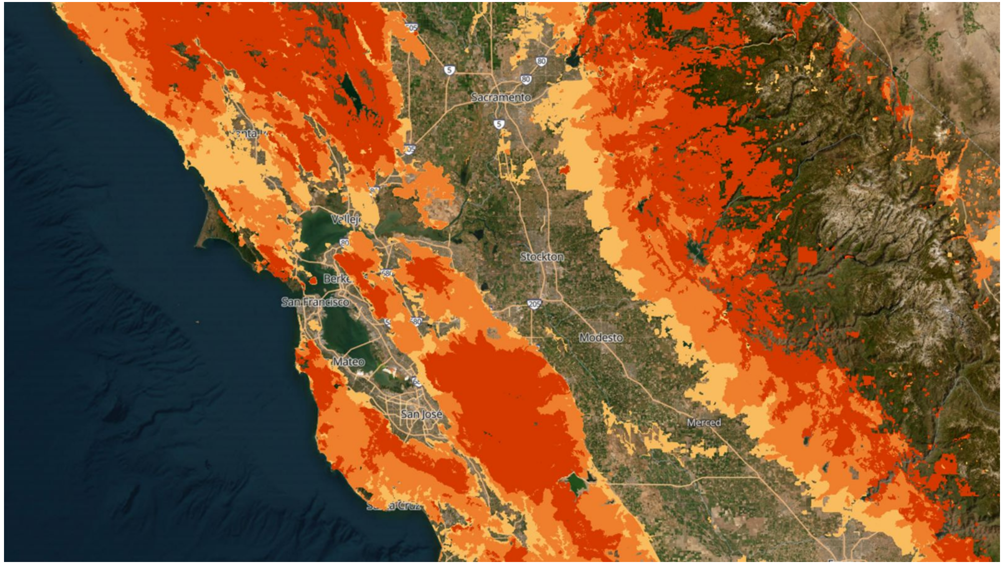
2. Airplane Design for Real-Time Data Collection

3. Translating Diverse Sensor Data into ML Inputs



LA Palisade Fires 2025 – Before and After

We're here!



Fire risk map for Central California

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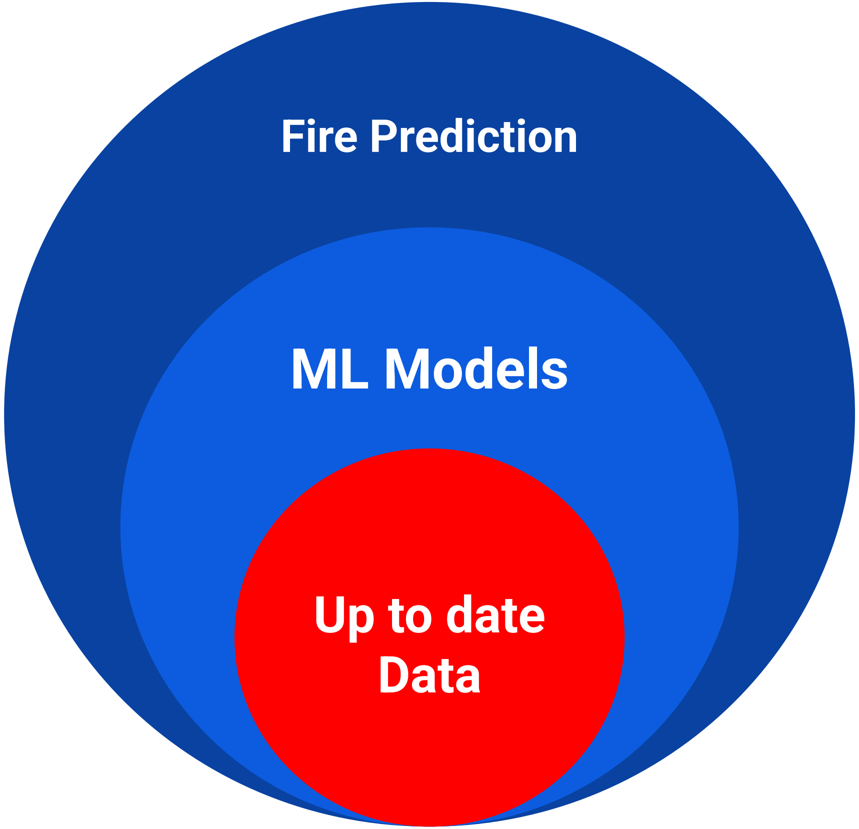

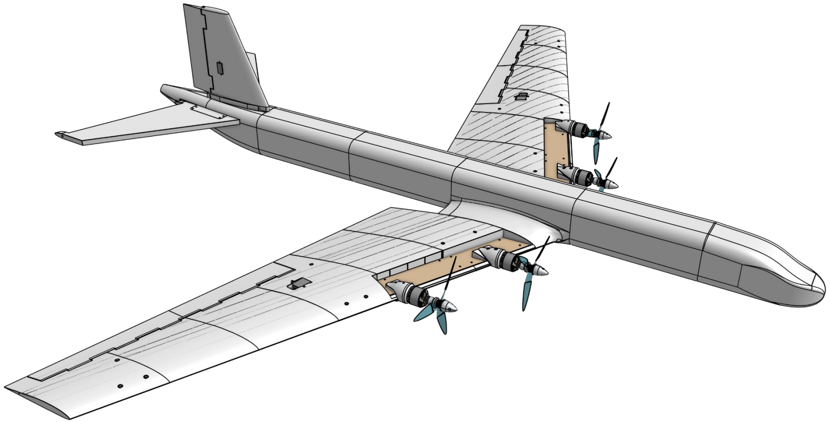

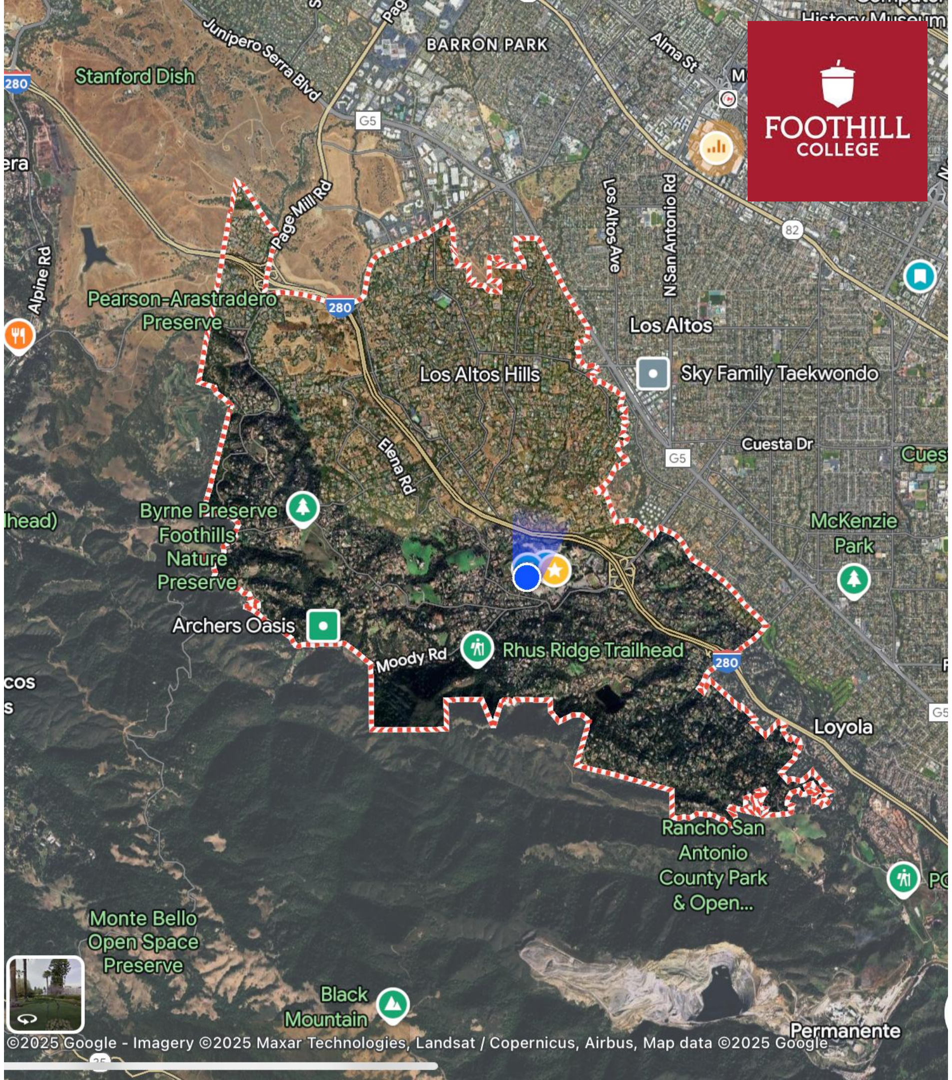

BARRON PARK
Junipero Serra Blvd
Alma St
Stanford Dish
280
G5
Page Mill Rd
Los Altos Ave
N San Antonio Rd
82
Alpine Rd
Pearson-Arastradero Preserve
Los Altos
Los Altos Hills
Sky Family Taekwondo
Cuesta Dr
Elena Rd
Byrne Preserve
Foothills Nature Preserve
McKenzie Park
Archers Oasis
Moody Rd
Rhus Ridge Trailhead
Loyola
Rancho San Antonio County Park & Open...
Monte Bello Open Space Preserve
Black Mountain
Permanente
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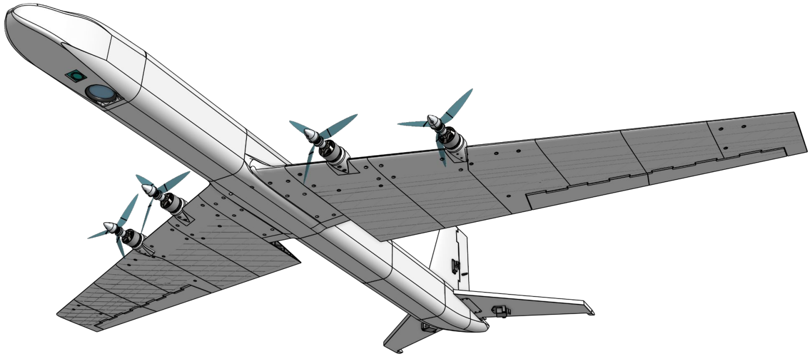

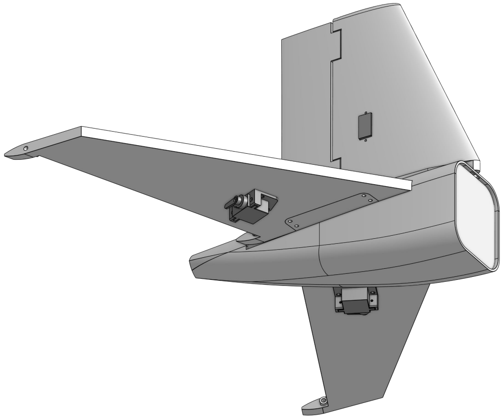

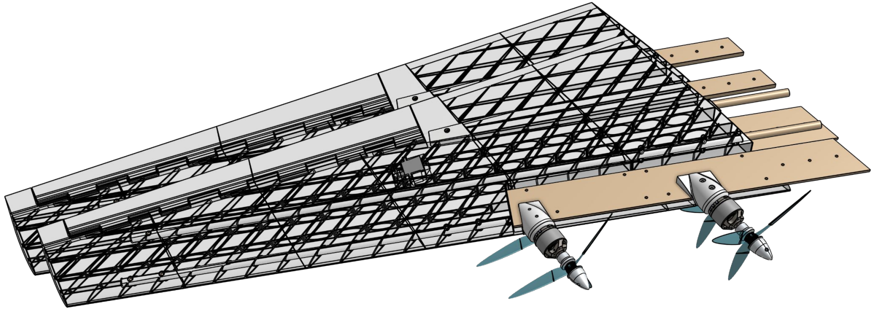

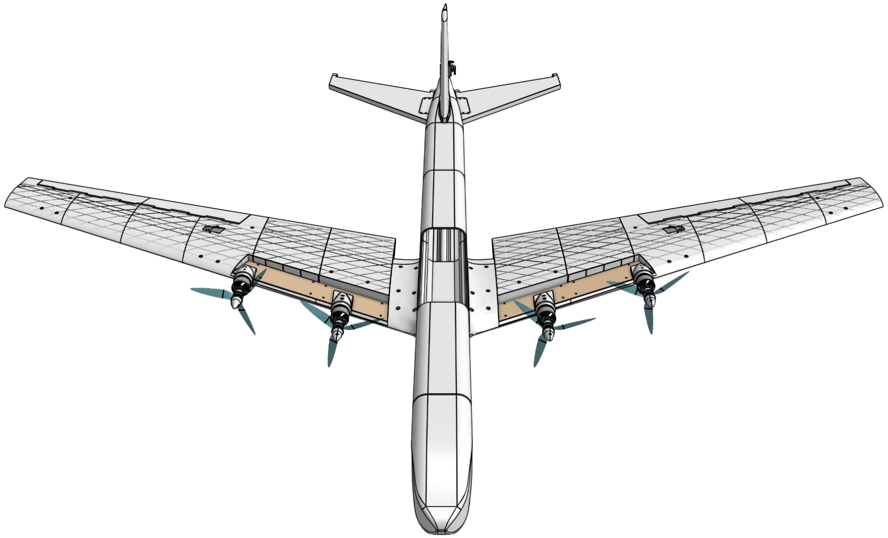

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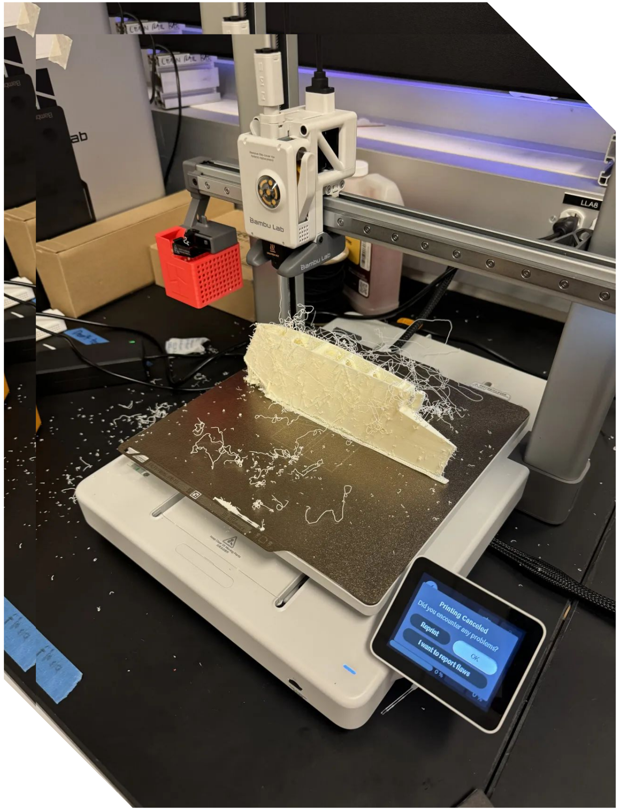

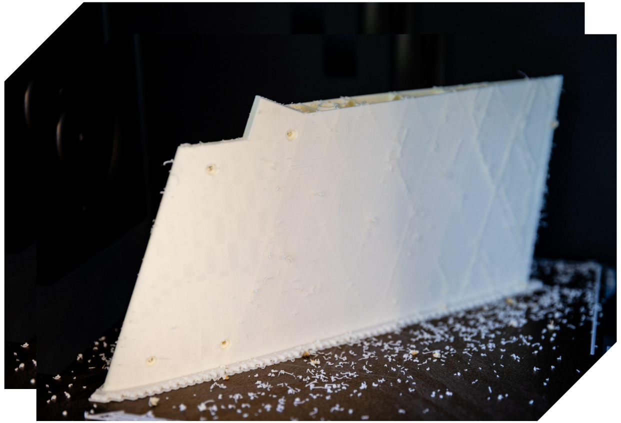

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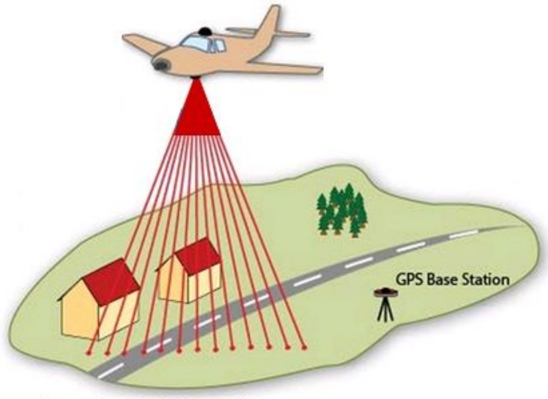
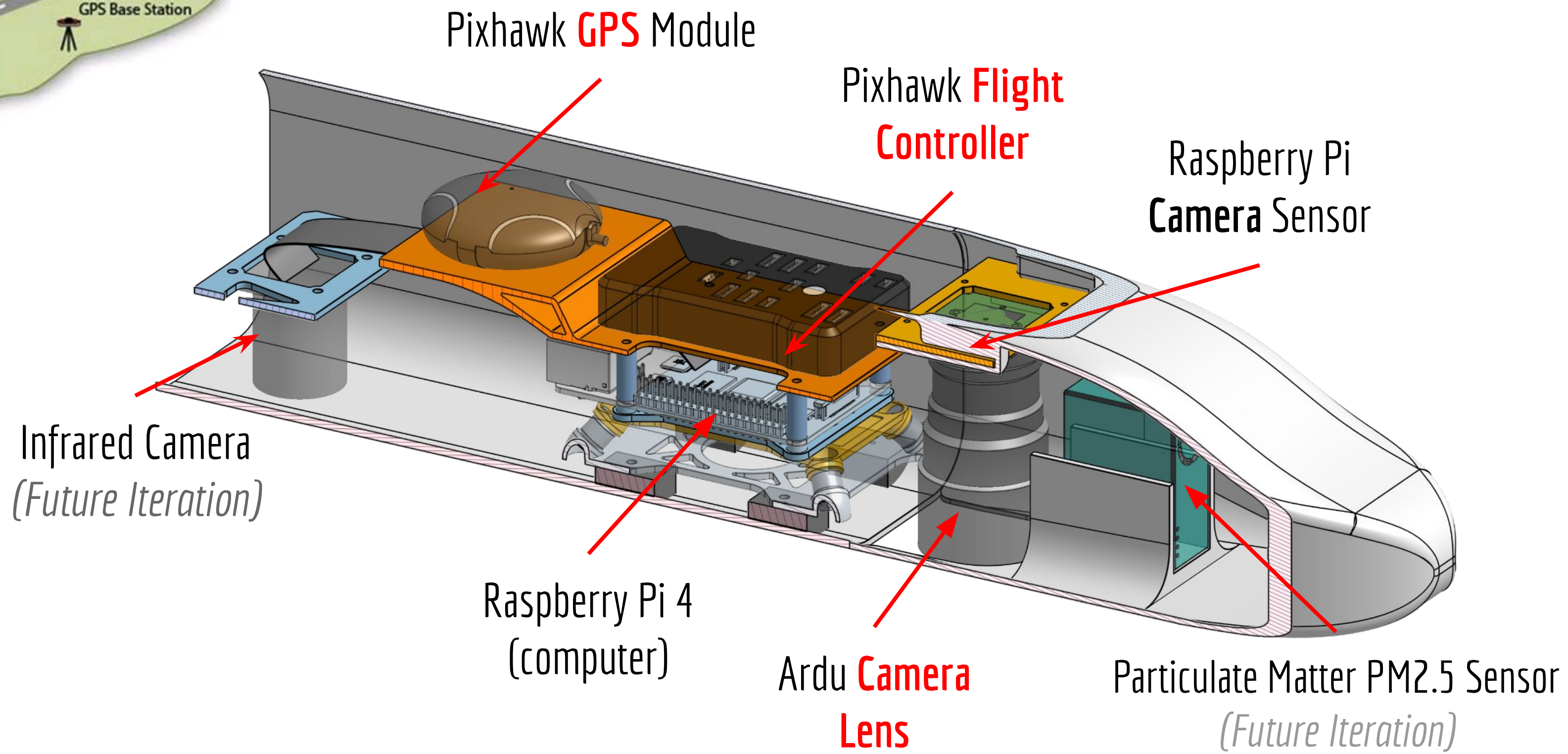
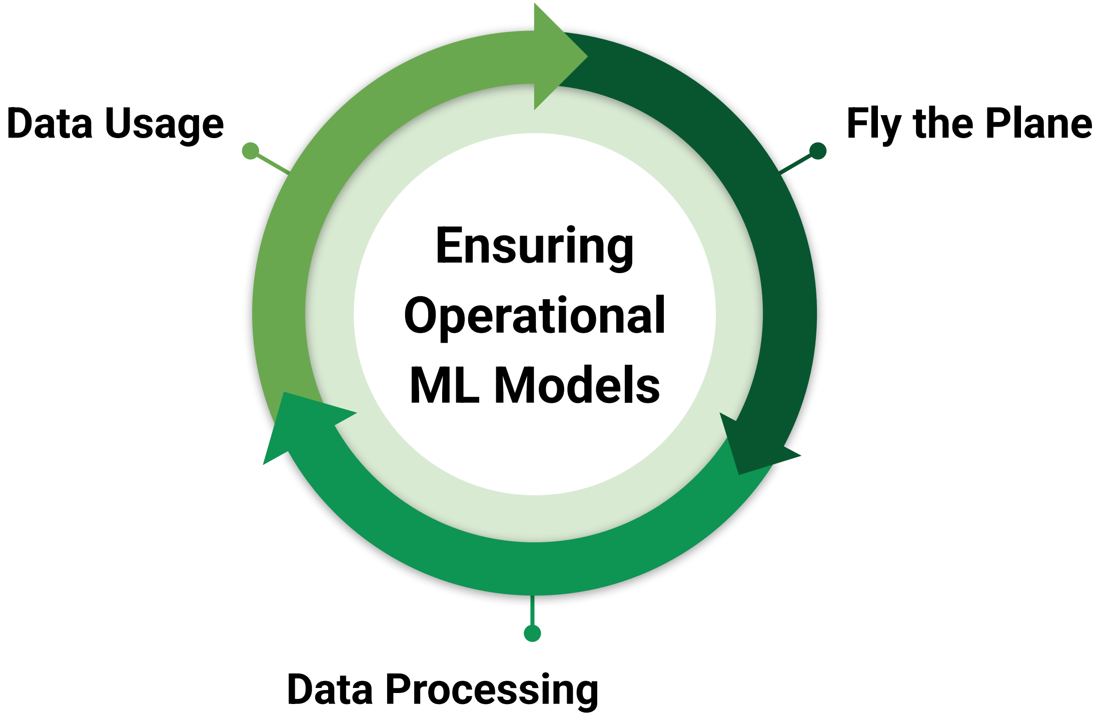

GPS Base Station


Pixhawk GPS Module
Pixhawk Flight Controller
Raspberry Pi Camera Sensor
Infrared Camera
(Future Iteration)
Raspberry Pi 4
(computer)
Ardu Camera Lens
Particulate Matter PM2.5 Sensor
(Future Iteration)


Data Usage
Fly the Plane
Ensuring Operational ML Models
Data Processing

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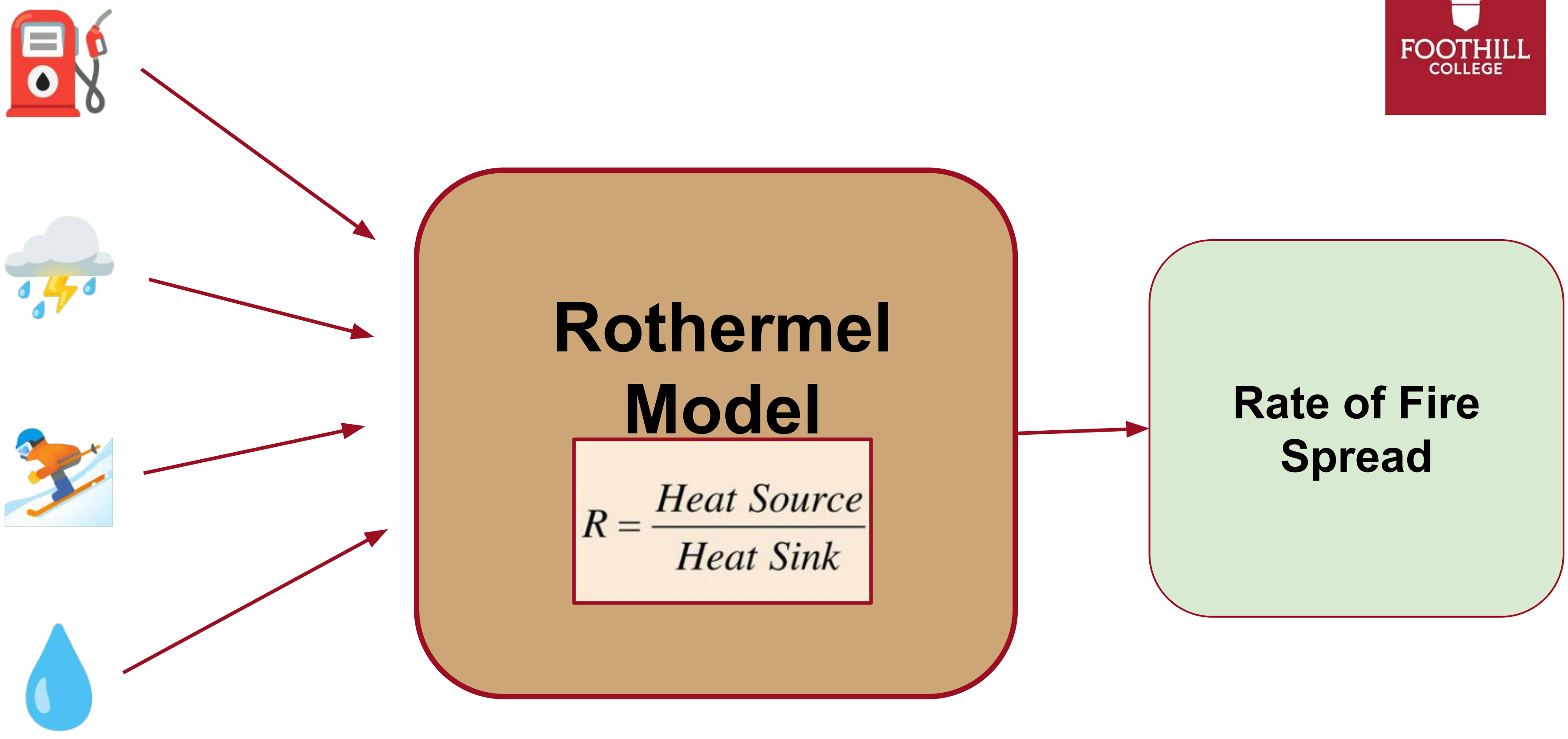
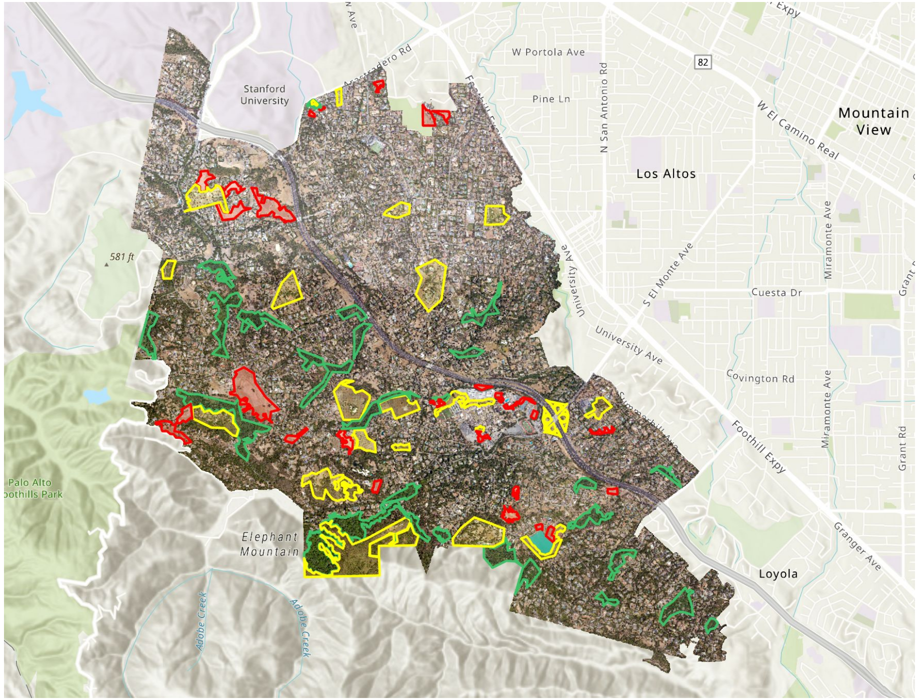


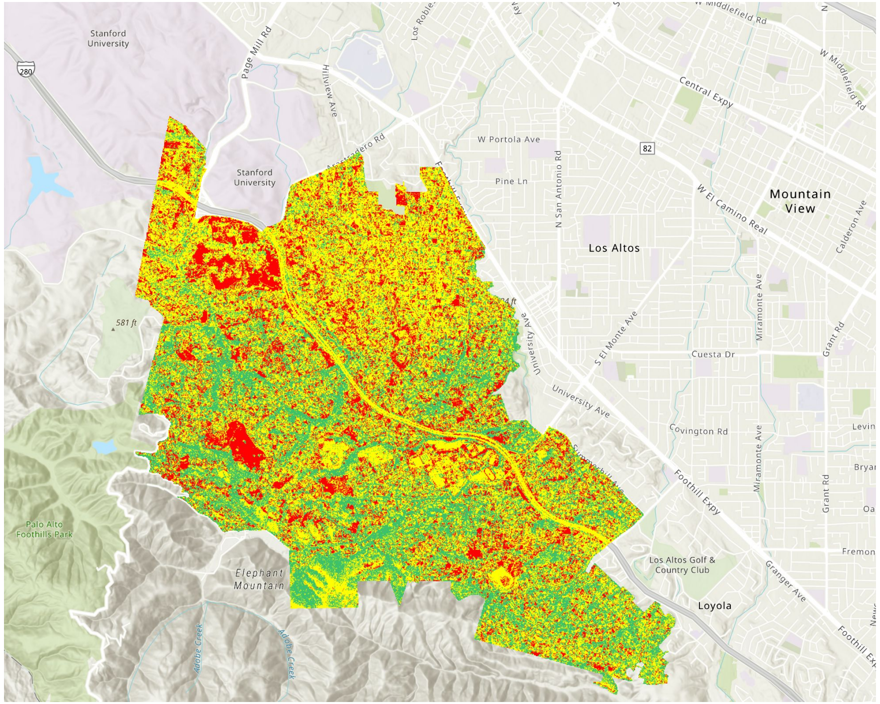
Dead vegetation vs live vegetation

[Dead Tree Removal For Wildfire Prevention: Our Guide | CAL FIRE]



	Risk Level
(green)	Low
(yellow)	Moderate
(red)	High

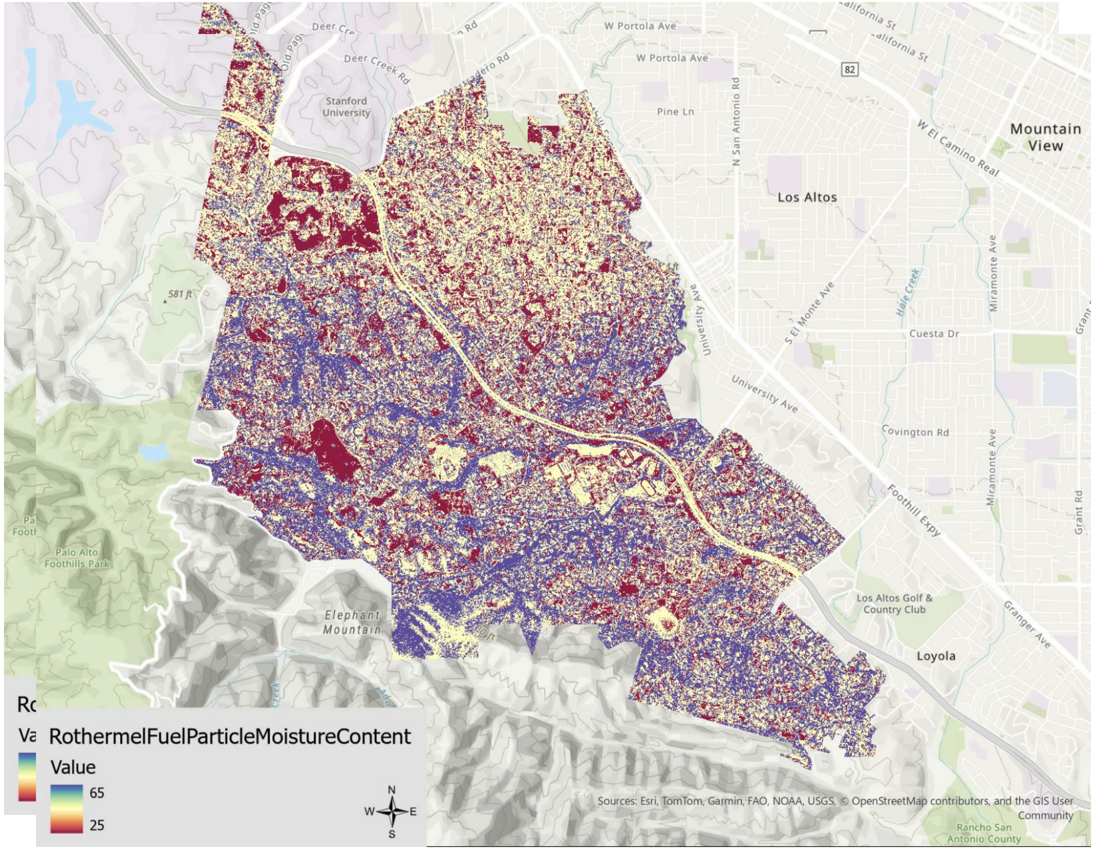
High quality training data for the three risk level categories



	Risk Level	Moisture Level %
	Low	65
	Moderate	45
	High	25

30%

Complete Image Classification for the Three Risk Level Categories



Range of moisture percentages for Los Altos Hills

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ETI

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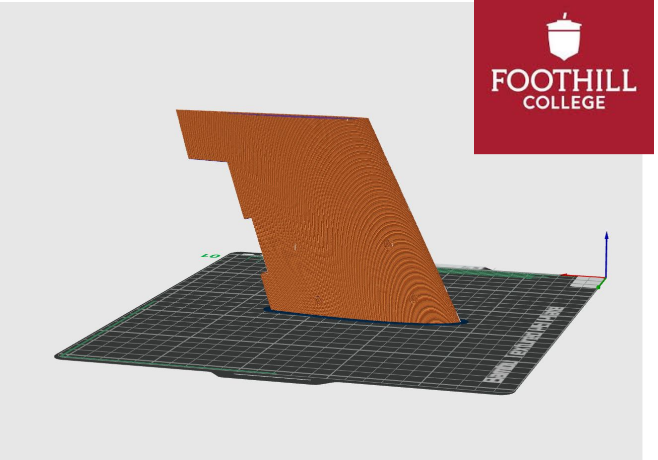

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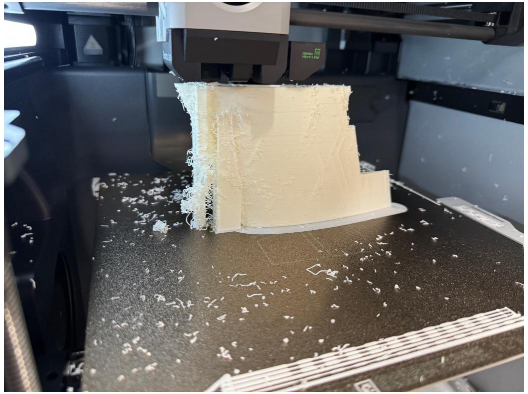
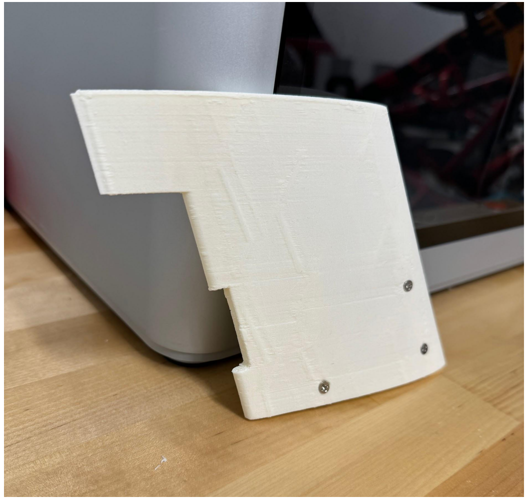
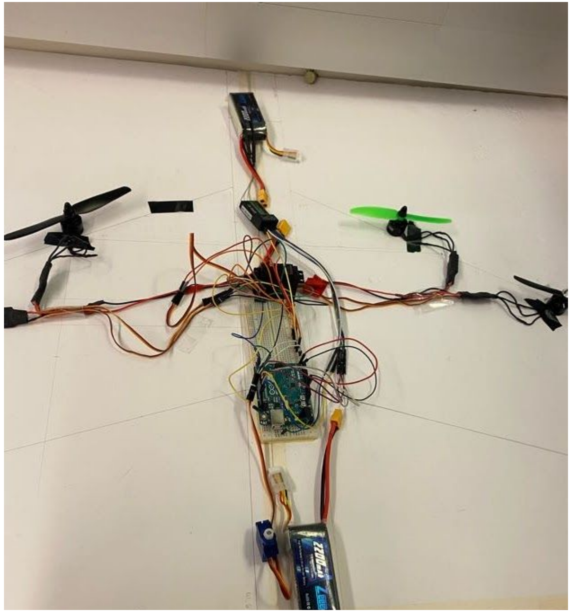
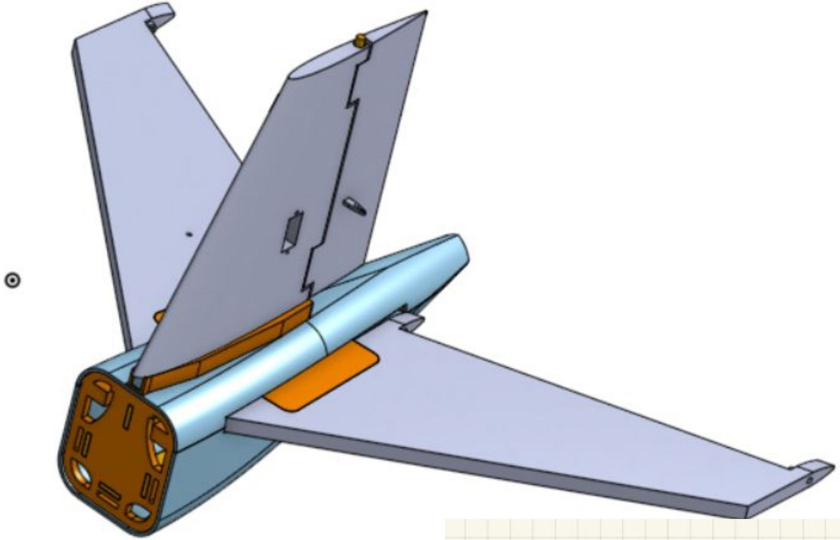


KCI

Makerspace







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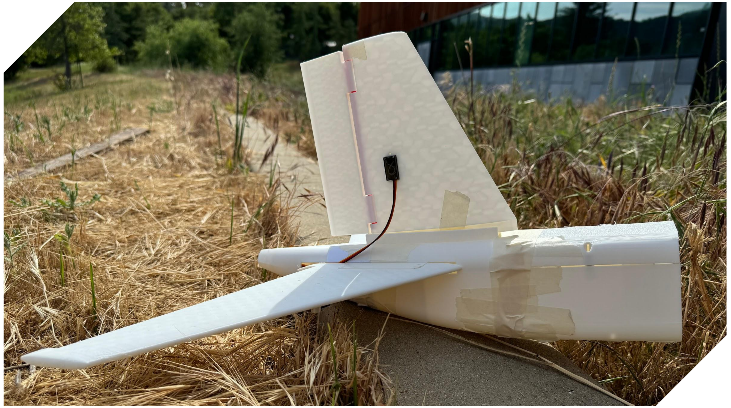

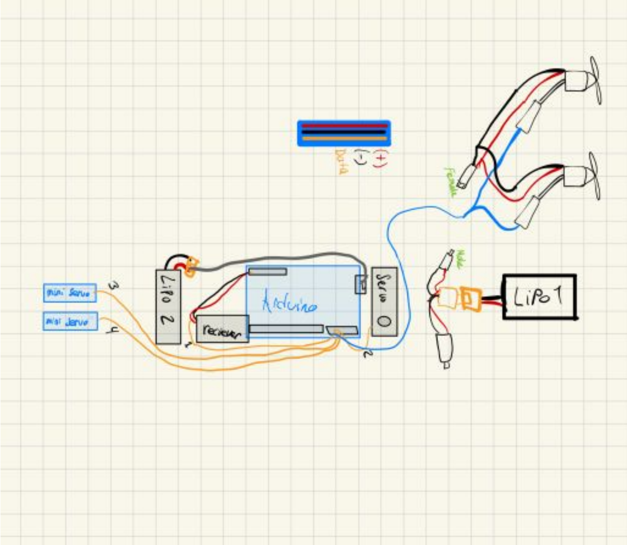

Thank you!